Supplementary Material

Label-Space Mismatch Distorts Cross-Cohort Evaluation in Breast Cancer Molecular Subtyping

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S1. Transformer Architecture Investigation

This supplementary section provides detailed results for the Transformer architecture investigation described in Section 7 of the main manuscript.

S1.1 Capacity Sweep

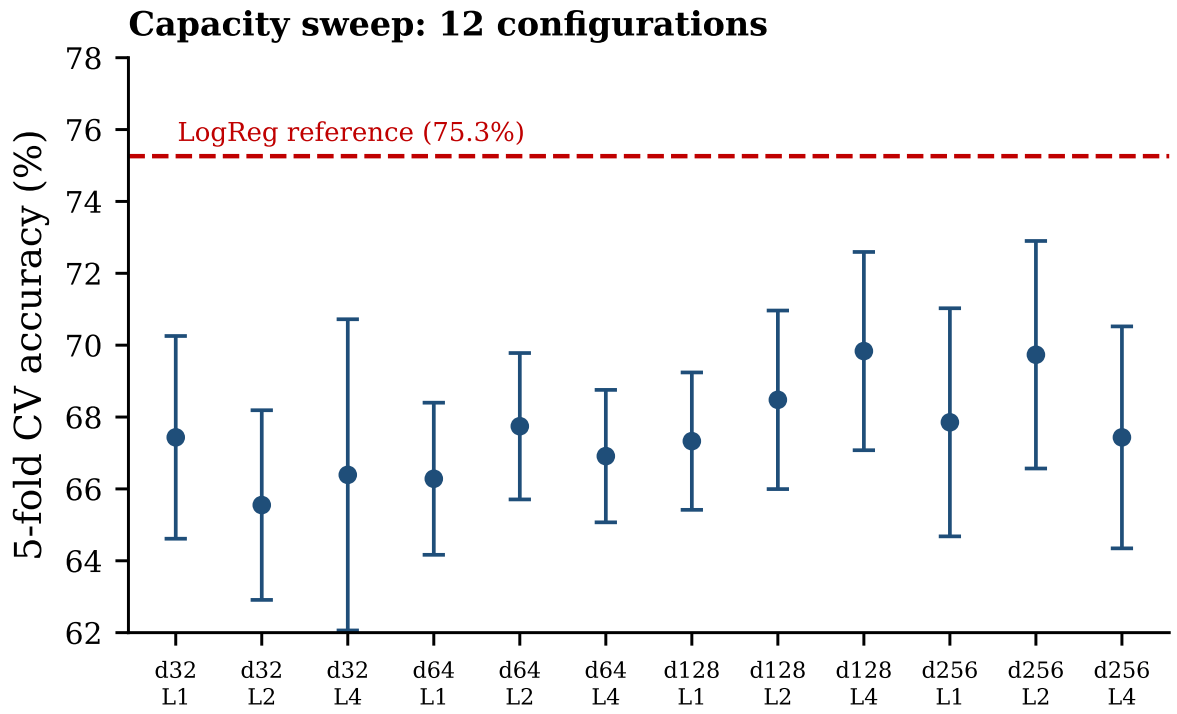


Figure 1: Capacity sweep over 12 configurations ($d \times L$). Every configuration lies below the logistic-regression reference (dashed), and eleven of twelve fall within a 4-point band, indicating that capacity is not the binding constraint.

S1.2 Learning Curves

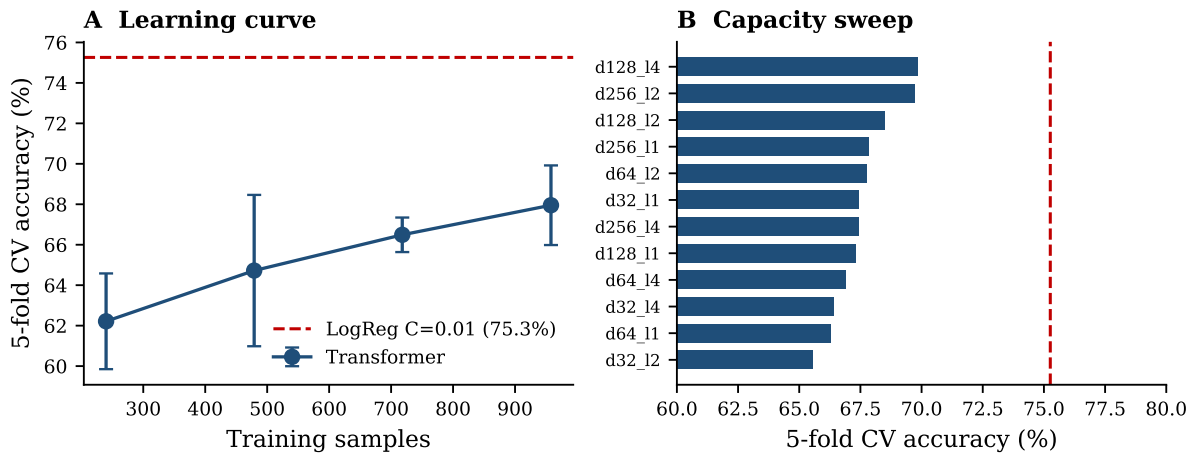


Figure 2: Learning curve for the best small configuration, with the logistic-regression reference. The curve is decelerating; extrapolation places parity at roughly six times the present training set.

S1.3 Regularization Sweep

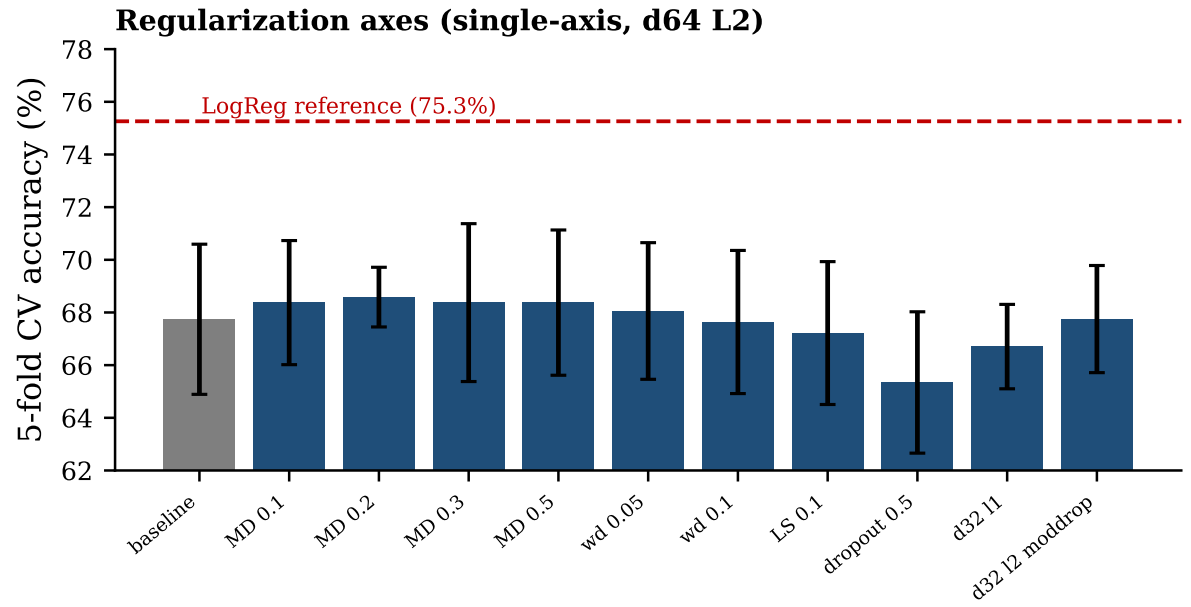


Figure 3: Single-axis regularization sweep. The strongest intervention gains 0.84 points, and no configuration approaches the logistic-regression reference.

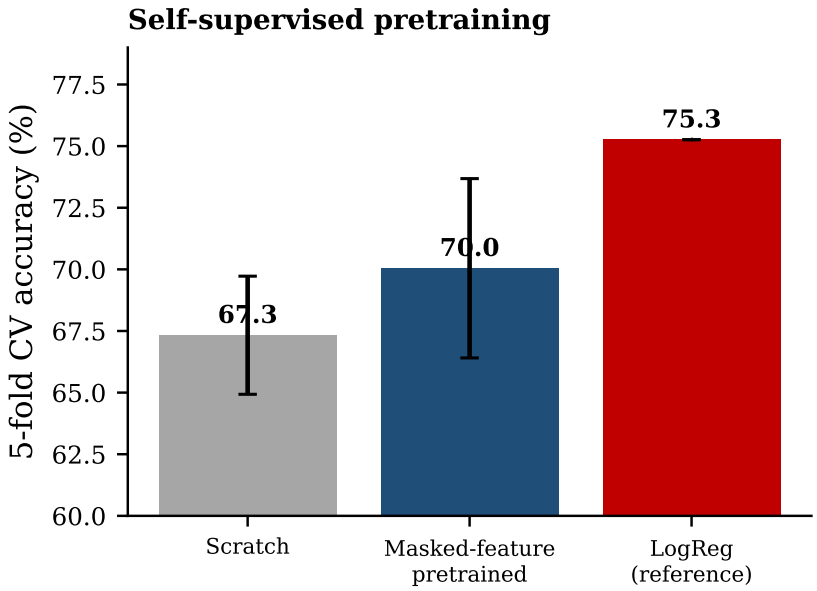


Figure 4: Masked-feature pretraining raises cross-validated accuracy by 2.7 points, the largest internal gain of any intervention tested, without reaching the linear reference.

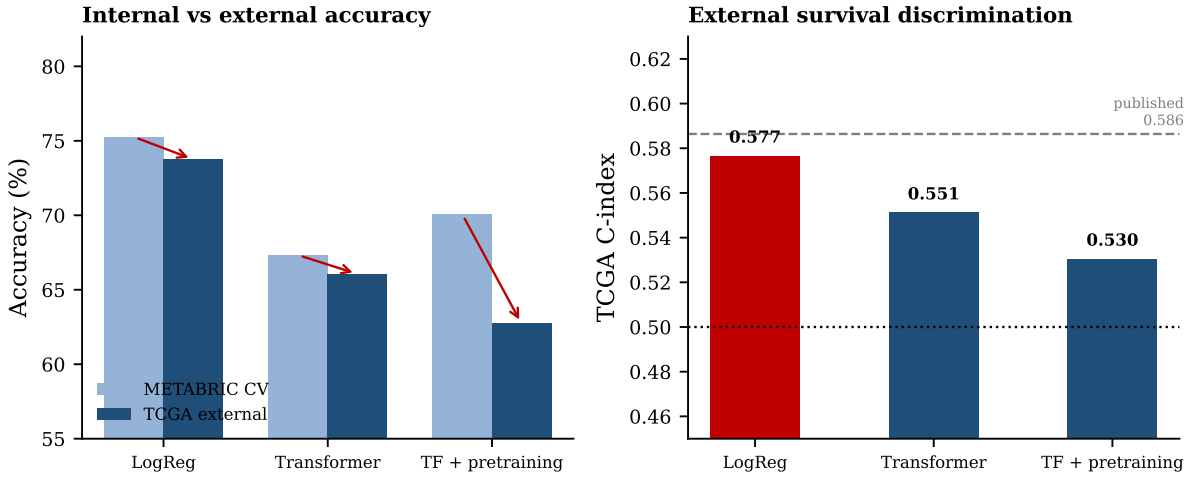


Figure 5: Internal versus external performance. Left: accuracy on METABRIC cross-validation and on TCGA. Right: external survival C-index. The pretrained Transformer shows the largest internal-to-external drop, and the lowest external C-index.

S1.4 Pretraining Results

S1.5 External Validation Reversal

S1.6 Supplementary Tables

Table 1: Complete feature engineering details for all five modalities.

Modality	**Variables**	**Encoding**	**Dimensions**
Clinical	22 variables	One-hot + Z-score	69
mRNA	412 genes	Z-score	412
CNA	256 genes	Z-score	256
Methylation	232 probes	Z-score	232
Mutation	173 genes	Binary	173
Total	**1,095**		**1,142**

Table 2: ANOVA feature selection results for mRNA genes.

Gene	**F-statistic**	**Rank**
ACOT4	130.8	1
A2ML1	102.0	2
ABAT	98.5	3
⋮	⋮	⋮